THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the insolation is 1 kW m^{-2}, the solar energy received by 1 m^2 in 1 h is: 1

(a) 2 kWh **(b)** 8 kWh **(c)** 4 kWh **(d)** 1 kWh

2. The solar constant is about 1400 J s^{-1} m^{-2}. The energy arriving on 2 m^2 (perpendicular to the Sun's rays) in 1 s is: 1

(a) 5600 J **(b)** 2800 J **(c)** 1400 J **(d)** 3800 J

3. A wavelength of 100 nm in metres is: 1

(a) 100 × 10^{-12} m **(b)** 100 × 10^{-3} m **(c)** 100 × 10^{-6} m **(d)** 100 × 10^{-9} m

4. The main source of energy for the Earth is: 1

(a) tidal energy **(b)** fossil fuels
(c) solar radiation **(d)** geothermal energy

5. Radiation that warms the Earth's surface and is re-radiated is mainly: 1

(a) infrared **(b)** gamma **(c)** X-rays **(d)** radio

6. The polar regions are colder than the equator mainly because: 1

(a) they have more oceans
(b) they have no atmosphere
(c) they are closer to the Sun
(d) the Sun's rays are spread over a larger area

7. Greenhouse gases include: 1

(a) neon and hydrogen **(b)** nitrogen and oxygen
(c) carbon dioxide, methane and water vapour **(d)** argon and helium

8. The agreement that reduced the use of CFCs is the: 1

(a) Paris Agreement **(b)** Montreal Protocol
(c) Rio Declaration **(d)** Kyoto Protocol

9. The surface that reflects most solar radiation is: 1

(a) forest **(b)** dark soil **(c)** fresh snow **(d)** ocean

10. Conversion of water vapour into droplets is: 1

(a) infiltration **(b)** evaporation **(c)** condensation **(d)** transpiration

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Name the four spheres of the Earth system. 2

12. Give an example of how a change in one sphere affects another. 2

13. What is solar radiation? Why is it important? 2

14. Differentiate between the solar constant and insolation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain how a change in the cryosphere can affect the other spheres. 3

16. Describe the electromagnetic spectrum from long to short wavelengths. 3

17. Why is UV radiation harmful and useful? 3

OR

Calculate the energy received by a 5 m^2 solar panel in 2 hours at 1 kW m^{-2} insolation. 3

18. Why is insolation at the surface less than the solar constant? 3

OR

Why is India suitable for solar energy? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Name and describe the four spheres of the Earth. 5
(ii) Explain with an example how they are interlinked.
(iii) What is the role of the cryosphere?

OR

Case study: Less snowfall in winter leads to less water in a lake in summer and less grass. 5
(i) Which spheres are involved?
(ii) Trace the chain of effects.
(iii) How might this affect animals?
(iv) Suggest one measure.

20. (i) What is solar radiation? 5
(ii) Describe the electromagnetic spectrum.
(iii) What are the roles of UV, visible and infrared radiation?

OR

Case study: A solar panel of 4 m^2 receives 1 kW m^{-2} sunlight for 5 hours. 5
(i) Find the power.
(ii) Find the energy in kWh.
(iii) Express it in joules.
(iv) What is the difference between the solar constant and this value?

END OF QUESTION PAPER — Total: 40 Marks